Intersection of Two Planes

2. Do the following pairs of planes intersect in a straight line?
   a. $-6x + 12y - 9z + 9 = 0$ and $4x - 8y + 6z + 9 = 0$
   b. $2x - y - 2z + 3 = 0$ and $6x - 3y - 6z + 9 = 0$
   c. $\vec{r} = (6, 0, 1) + p(1, 1, 2) + q(4, 2, 3)$
      and $\vec{r} = (1, 1, -9) + s(5, 3, 5) + t(3, 1, 1)$
   d. $\vec{r} = (1, 1, 1) + p(0, 0, 1) + q(0, 1, 0)$
      and $\vec{r} = (0, 0, 0) + s(0, 0, 1) + t(1, 0, 0)$

3. Determine which of the following pairs of planes are parallel. For each pair that is not parallel, find the parametric equations of the line of intersection. Use algebraic elimination.
   a. $x + y - 3z = 4$ and $x + 2y - z = 1$
   b. $5x - 2y + 2z + 1 = 0$ and $5x - 2y + 2z - 3 = 0$
   c. $x - 3y - z + 3 = 0$ and $2x + 4y - z - 5 = 0$
   d. $x + y + z = 1$ and $x = 0$
   e. $x + 3y - z - 4 = 0$ and $2x + 6y - 2z - 8 = 0$

4. Write the augmented matrix for each of the following linear systems.
   a. $3x - 7y + z = 12$
      $x + y - 2z = -3$
   b. $-4x + 3y + 2z = 4$
      $2y - 5z = 5$
   c. $x + 4z = 16$
      $y - 8z = -2$
   d. $5y - 2z + 6x = 4$
      $3z + 5y - 2x = -4$

5. Write the system of equations that corresponds to each of these matrices.
   a. $\left[\begin{array}{ccc|c} 1 & 0 & 4 & 9 \\ 0 & 1 & -6 & 4 \end{array}\right]$
   b. $\left[\begin{array}{ccc|c} 8 & -2 & 3 & -6 \\ 2 & -6 & -6 & 9 \end{array}\right]$
   c. $\left[\begin{array}{ccc|c} 5 & 0 & -10 & 8 \\ 0 & 3 & -4 & 6 \end{array}\right]$
   d. $\left[\begin{array}{ccc|c} 1 & 0 & 4 & 0 \\ 0 & 1 & 9 & 0 \end{array}\right]$

6. Use Gauss-Jordan elimination to find the vector equation of the line of intersection of each pair of planes.

a. $x + 2y + 7z = 4$
$x + 3y + 3z = 1$

b. $x - 4y + 3z = 5$
$2x + y + 6z = 0$

c. $2x + 8y + 2z = 7$
$x + 4y - z = 3$

d. $4x - 8y - 3z = 6$
$-3x + 6y + z = -2$

e. $3x + 2y - 6z = 5$
$2x + 3y - 9z = -10$

f. $6x + 8y - 3z = 9$
$10x - 2y - 5z = 15$

Answers:

2. a. no **b.** no **c.** no **d.** yes **3. a.** $x = 7 + 5t, y = -3 - 2t, z = t$ **b.** parallel

c. $x = \frac{3+7t}{10}, y = \frac{11-t}{10}, z = t$

d. $x = 0, y = 1 - t, z = t$ **e.** parallel

4. a. $\left[\begin{array}{ccc|c} 3 & -7 & 1 & 12 \\ 1 & 1 & -2 & -3 \end{array}\right]$ **b.** $\left[\begin{array}{ccc|c} -4 & 3 & 2 & 4 \\ 0 & 2 & -5 & 5 \end{array}\right]$

c. $\left[\begin{array}{ccc|c} 1 & 0 & 4 & 16 \\ 0 & 1 & -8 & -2 \end{array}\right]$ **d.** $\left[\begin{array}{ccc|c} 6 & 5 & -2 & 4 \\ -2 & 5 & 3 & -4 \end{array}\right]$

5. a. $x + 4z = 9, y - 6z = 4$

b. $8x - 2y + 3z = -6, 2x - 6y - 6z = 9$

c. $5x - 10z = 8$, $3y - 4z = 6$ **d.** $x + 4z = 0, y + 9z = 0$

6. a. $\vec{r} = (10, -3, 0) + t(-15, 4, 1)$

b. $\vec{r} = \left(\frac{5}{9}, \frac{-10}{9}, 0\right) + t(-3, 0, 1)$

c. $\vec{r} = \left(\frac{13}{4}, 0, \frac{1}{4}\right) + t(-4, 1, 0)$

d. $\vec{r} = (0, 0, -2) + t(2, 1, 0)$

e. $\vec{r} = (7, -8, 0) + t(0, 3, 1)$ **f.** $\vec{r} = \left(\frac{3}{2},0,0\right) + t\left(\frac{1}{2},0,1\right)$